CS5487: Machine Learning: Principles and Practice

Assignment 2

Semester A, 2026/27

Name: _____

Student ID: _____

Total: 100 marks

Academic Integrity: Individual Work Only

- **This assignment must be completed independently.** Do not collaborate with others, copy answers, or share solutions.
- **The use of AI tools is strictly prohibited.** Do not use ChatGPT, Claude, Gemini, Copilot, or any other AI assistant for solving questions, deriving results, checking calculations, or writing or revising your answers. All submitted work must be your own.

Submission Requirements

- Submit **one PDF file** containing your answers to all six questions.
- Typed work should be exported to PDF. For handwritten work, **scan every page** with a scanner or document-scanning app, then combine the scans into one clearly legible PDF.
- **Direct photographs are not accepted.** Do not submit ordinary phone photos or photos inserted into a PDF. Check that scanned pages are clear, correctly oriented, and in order.

Assessment Instructions

- Six questions: one calculation and one derivation problem each from Weeks 5, 7, and 8.
- Show essential reasoning and intermediate steps. Numerical answers without supporting work may receive limited credit.
- A calculator may be used. Unless stated otherwise, report numerical approximations to three decimal places.

Question	1	2	3	4	5	6	Total
Marks	15	15	20	15	15	20	100

Question 1: Spectral Clustering from Eigenvectors [15 marks]

An undirected graph has four observations as nodes. Its similarity matrix is

$$\boldsymbol{W} = \begin{bmatrix} 0 & 4 & 1 & 0 \\ 4 & 0 & 0 & 1 \\ 1 & 0 & 0 & 4 \\ 0 & 1 & 4 & 0 \end{bmatrix}.$$

Here w_{ij} is the strength of the connection between nodes i and j. Use the unnormalized graph Laplacian $\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{W}$, where $\boldsymbol{D}$ contains the weighted node degrees.

Consider these four vectors:

$$\boldsymbol{u}_0 = (1,1,1,1)^\top, \qquad \boldsymbol{u}_1 = (1,1,-1,-1)^\top,$$
$$\boldsymbol{u}_2 = (1,-1,1,-1)^\top, \quad \boldsymbol{u}_3 = (1,-1,-1,1)^\top.$$

(a) Compute the degree matrix $\boldsymbol{D}$ and the Laplacian $\boldsymbol{L}$. [3 marks]

(b) Verify that each $\boldsymbol{u}_j$ is an eigenvector of $\boldsymbol{L}$ and compute its eigenvalue. [6 marks]

(c) Identify the eigenvector associated with the smallest *nonzero* eigenvalue. Assign the four nodes to two clusters using the signs of its entries. [4 marks]

(d) Explain why the resulting groups agree with the graph connection strengths. [2 marks]

Question 2: Mean Shift as Density Ascent [15 marks]

For observations $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$, consider a Gaussian kernel density estimate

$$\hat{p}_h(\boldsymbol{x}) = \frac{1}{n(2\pi)^{d/2}h^d} \sum_{i=1}^{n} \exp\left(-\frac{\|\boldsymbol{x} - \boldsymbol{x}_i\|_2^2}{2h^2}\right), \qquad h > 0.$$

Define the unnormalized weights and their weighted mean as

$$w_i(\boldsymbol{x}) = \exp\left(-\frac{\|\boldsymbol{x} - \boldsymbol{x}_i\|_2^2}{2h^2}\right), \qquad \boldsymbol{m}_h(\boldsymbol{x}) = \frac{\sum_i w_i(\boldsymbol{x})\boldsymbol{x}_i}{\sum_i w_i(\boldsymbol{x})}.$$

You may assume $\hat{p}_h(\boldsymbol{x}) > 0$ for every $\boldsymbol{x}$.

(a) Differentiate $w_i(\boldsymbol{x})$ with respect to $\boldsymbol{x}$ to obtain $\nabla_{\boldsymbol{x}} w_i(\boldsymbol{x})$. [4 marks]

(b) Starting from the KDE, derive

$$\nabla_{\boldsymbol{x}} \log \hat{p}_h(\boldsymbol{x}) = \frac{\boldsymbol{m}_h(\boldsymbol{x}) - \boldsymbol{x}}{h^2}.$$

Show the algebra connecting the derivative and the weighted mean. [7 marks]

(c) Use the result to express one Gaussian mean-shift update as a step along the log-density gradient. Explain what holds at a fixed point and why a stationary point need not be a density maximum. [4 marks]

Question 3: Gaussian QDA and Decision Costs [20 marks]

A two-class classifier models the features $\boldsymbol{x} = (x_1, x_2)^\top$ by

$$\boldsymbol{x} \mid y = 0 \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \right), \quad \pi_0 = 0.6,$$
$$\boldsymbol{x} \mid y = 1 \sim \mathcal{N}\left(\begin{bmatrix} 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \right), \quad \pi_1 = 0.4.$$

Under equal misclassification costs, compare Gaussian scores

$$g_k(\boldsymbol{x}) = -\tfrac{1}{2} d_k^2(\boldsymbol{x}) - \tfrac{1}{2} \log |\boldsymbol{\Sigma}_k| + \log \pi_k, \qquad d_k^2(\boldsymbol{x}) = (\boldsymbol{x} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_k).$$

(a) Compute both covariance inverses and determinants, then express $d_0^2(\boldsymbol{x})$ and $d_1^2(\boldsymbol{x})$ in terms of x_1, x_2. [5 marks]

(b) Compute $g_1 - g_0$ at $\boldsymbol{x}_A = (0, 0)^\top$ and $\boldsymbol{x}_B = (2, 2)^\top$. Classify both under equal costs. [5 marks]

(c) Derive the explicit equation of the decision boundary $g_1(\boldsymbol{x}) = g_0(\boldsymbol{x})$. Identify the terms that make it quadratic. [6 marks]

(d) For $\boldsymbol{x}_M = (1, 1)^\top$, compute $P(y = 1 \mid \boldsymbol{x}_M)$ and compare the zero–one-loss decision with the minimum-risk decision when $C_{\mathrm{FN}} = 2$ and $C_{\mathrm{FP}} = 1$. [4 marks]

Question 4: Bernoulli Naive Bayes as a Linear Classifier [15 marks]

A binary classifier has labels $y \in \{0, 1\}$ and binary features $\boldsymbol{x} = (x_1, \ldots, x_d)^\top$, where $x_j \in \{0, 1\}$. Under the Bernoulli Naive Bayes assumption,

$$p(\boldsymbol{x} \mid y = k) = \prod_{j=1}^{d} \theta_{jk}^{x_j} (1 - \theta_{jk})^{1 - x_j}, \qquad \theta_{jk} = P(x_j = 1 \mid y = k).$$

Assume $0 < \theta_{jk} < 1$ and class priors $\pi_k = P(y = k) > 0$.

(a) Starting from Bayes' rule, derive an expression for the *log posterior odds*

$$\log \frac{P(y = 1 \mid \boldsymbol{x})}{P(y = 0 \mid \boldsymbol{x})}$$

involving the prior odds and the Bernoulli feature probabilities. [4 marks]

(b) Rearrange your expression into the linear-score form

$$\boxed{\log \frac{P(y = 1 \mid \boldsymbol{x})}{P(y = 0 \mid \boldsymbol{x})} = b + \sum_{j=1}^{d} w_j x_j.}$$

Derive explicit formulas for b and every w_j. [8 marks]

(c) Explain what a positive w_j means when feature j changes from 0 to 1. Explain which part of the score changes if only the class priors change. [3 marks]

Question 5: PCA with Feature Standardization [15 marks]

Two measurement channels have different scales. The training mean and covariance are

$$\bar{\boldsymbol{x}} = \begin{bmatrix} 40 \\ 10 \end{bmatrix}, \qquad \boldsymbol{S} = \begin{bmatrix} 16 & 4 \\ 4 & 4 \end{bmatrix}.$$

The training standard deviations are 4 and 2. Standardize a point using

$$\boldsymbol{u} = \boldsymbol{A}(\boldsymbol{x} - \bar{\boldsymbol{x}}), \qquad \boldsymbol{A} = \mathrm{diag}(1/4, 1/2).$$

Perform PCA on the standardized training data, with covariance $\boldsymbol{R} = \boldsymbol{A}\boldsymbol{S}\boldsymbol{A}^\top$.

(a) Compute the standardized covariance $\boldsymbol{R}$. [3 marks]

(b) Find the eigenpairs of $\boldsymbol{R}$, identify PC1, and compute its explained-variance ratio. [4 marks]

(c) For $\boldsymbol{x}_* = (44, 10)^\top$, compute its PC1 coordinate and rank-one reconstruction in the original units. [5 marks]

(d) Compute the squared residual errors in standardized and original units. Explain the difference. [3 marks]

Question 6: PCA as Optimal Linear Reconstruction [20 marks]

Let centered observations $\tilde{\boldsymbol{x}}_1, \ldots, \tilde{\boldsymbol{x}}_n \in \mathbb{R}^d$ have covariance

$$\boldsymbol{S} = \frac{1}{n} \sum_{i=1}^{n} \tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_i^\top .$$

Choose r orthonormal directions as the columns of $\boldsymbol{V}_r \in \mathbb{R}^{d \times r}$, with $\boldsymbol{V}_r^\top \boldsymbol{V}_r = \boldsymbol{I}_r$. Define $\boldsymbol{P} = \boldsymbol{V}_r \boldsymbol{V}_r^\top$ and the reconstruction $\hat{\boldsymbol{x}}_i = \boldsymbol{P} \tilde{\boldsymbol{x}}_i$.

(a) Show that $\boldsymbol{P}^\top = \boldsymbol{P}$ and $\boldsymbol{P}^2 = \boldsymbol{P}$. Use these facts to show that a reconstruction and its residual are orthogonal. [4 marks]

(b) Starting from the average squared reconstruction error

$$E(\boldsymbol{V}_r) = \frac{1}{n} \sum_{i} \|\tilde{\boldsymbol{x}}_i - \hat{\boldsymbol{x}}_i\|_2^2,$$

derive $E(\boldsymbol{V}_r) = \mathrm{tr}(\boldsymbol{S}) - \mathrm{tr}(\boldsymbol{V}_r^\top \boldsymbol{S} \boldsymbol{V}_r)$. [7 marks]

(c) Given eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$, justify why the leading r eigenvectors minimize E and derive $E_{\min} = \sum_{j=r+1}^{d} \lambda_j$. [6 marks]

(d) For eigenvalues $(6, 2, 1, 0.5)$, find the smallest r giving $E \leq 1.6$. [3 marks]